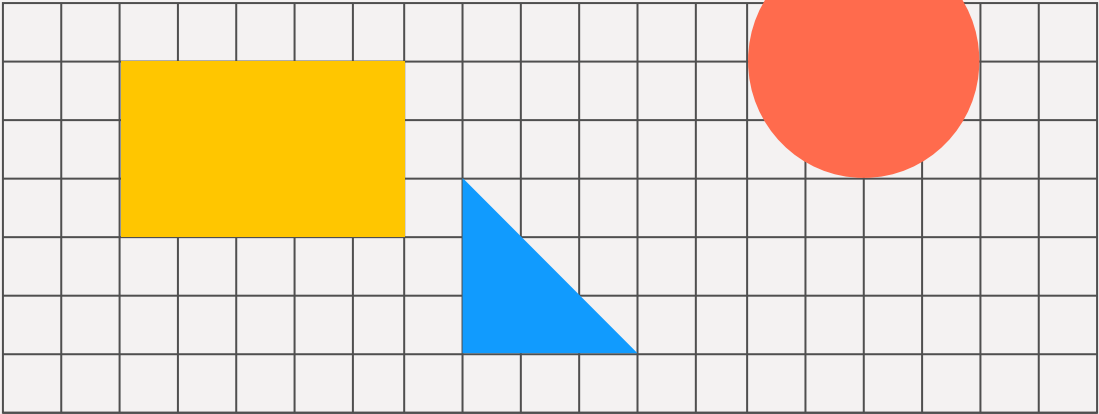
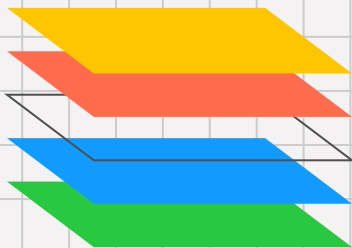
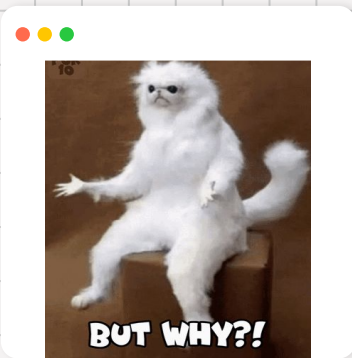


Modeling and Managing Tail Risk in Financial Returns

By Phan Nguyen Huong Le



Why fat... | tails



Goal

Quantify tail risk and simulate potential extreme losses to inform capital buffers and risk policies.

- **Tail risk** refers to the risk of rare, extreme losses in financial markets — events that lie in the "tails" of return distribution
- Extreme losses can destabilize portfolios.
- Standard models (e.g., normal distribution) **underestimate tail risk.**

Why focus on XLF?

Data

- Initially explored multiple assets (e.g., KO, PEP, VZ, PG) for model comparison
- Narrowed scope to XLF for depth over breadth:
 - XLF represents a sector **highly sensitive to tail risk** (Real-world crises (2008, COVID crash, SVB collapse) show that extreme events are more common than assumed)
 - Easier to interpret and evaluate model fit and performance
- Ensured robust model diagnostics and simulation on this single asset to showcase method effectiveness.

Baseline models: understand Static VaR & ES

Historical

- Based on empirical quantiles from historical returns.
- **No distributional assumption.**
- Pros: Non-parametric, flexible
- Cons:
 - Backward-looking, doesn't account for rare unseen events.

Parametric Normal

- Assumes returns are **normally** distributed
- VaR and ES derived analytically.
- Pros: Simple, fast
- Cons:
 - **Underestimates extreme losses.**
 - Thin tails; doesn't capture market shocks.

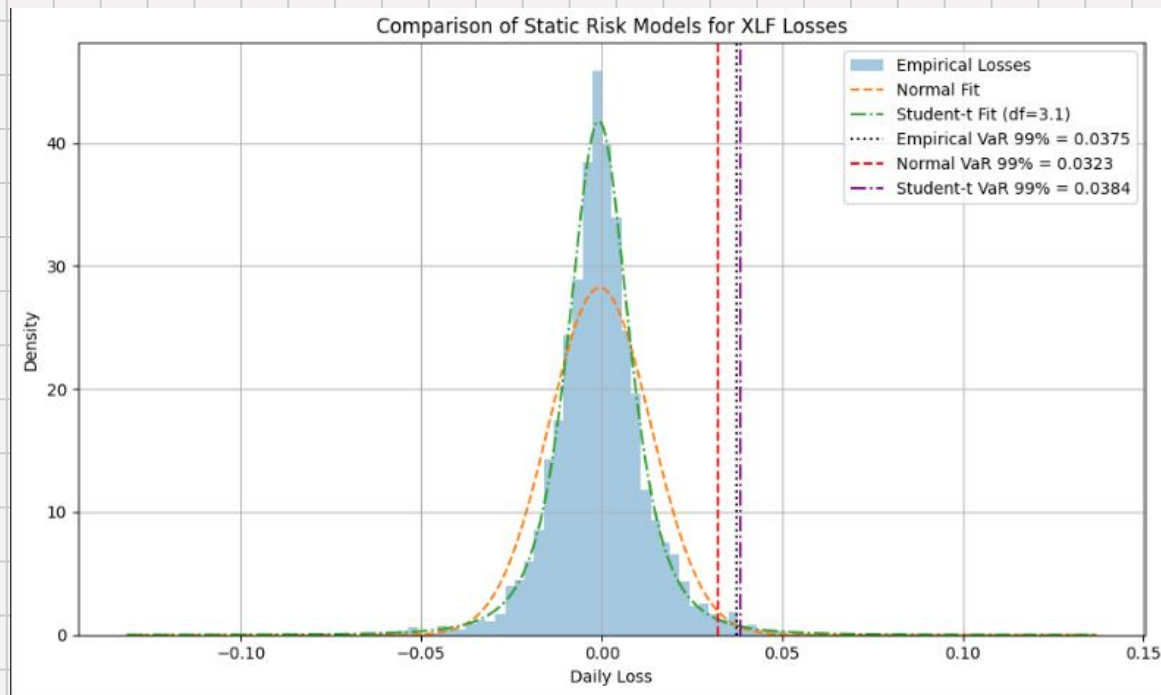
Parametric (t-dist)

- Returns follow a **t-distribution**, allowing fatter tails
- Better accounts for extreme observations than Normal.
- Pros: Captures tail risk better
- Cons: Still assumes symmetry Assumes i.i.d., no time-varying behavior.

Main results

Baseline Models on XLF

	Model	VaR 99%	ES 99%
0	Empirical	0.0375	0.0559
1	Normal	0.0323	0.0371
2	Student-t	0.0384	0.0590



Why Static Models **Fail**

Baseline Models

- Assume i.i.d. returns — ignores volatility clustering and temporal dependence.
- Ignore fat tails — underestimate the likelihood of rare but severe losses.
- Provide false confidence — VaR under normal assumptions can look acceptable until it's catastrophically wrong.

Beyond static models

EVT & GPD



EVT (extreme value theorem):

- Focuses only on the tail — where extreme losses live
- Models exceedances beyond a high threshold (e.g., top 5%)

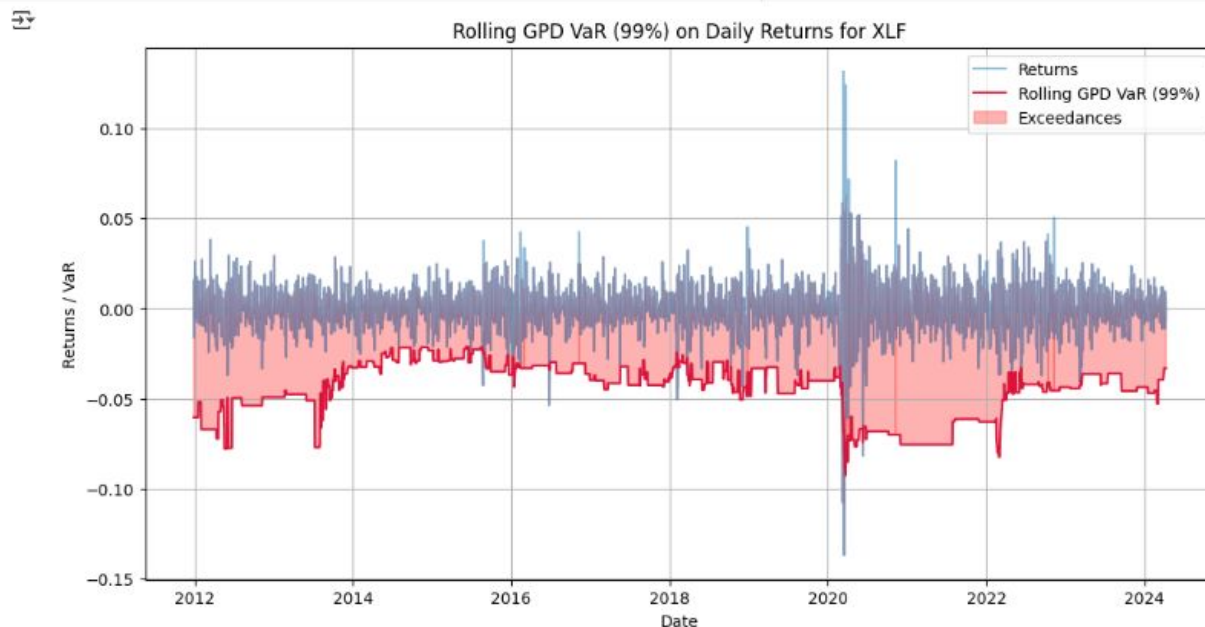
GPD (generalized pareto distribution)

- The go-to model under EVT for tail behavior
- Two parameters:
 - Shape (ξ) — how fat the tail is
 - Scale (β) — how spread out the extremes are

Main results

Rolling GPD on returns

📌 Kupiec POF Test for Rolling GPD VaR on Returns
Total Observations : 3089
Expected Violations (1%): 30.89
Actual Violations : 18
Observed Violation Rate : 0.0058
LR Statistic : 6.3921

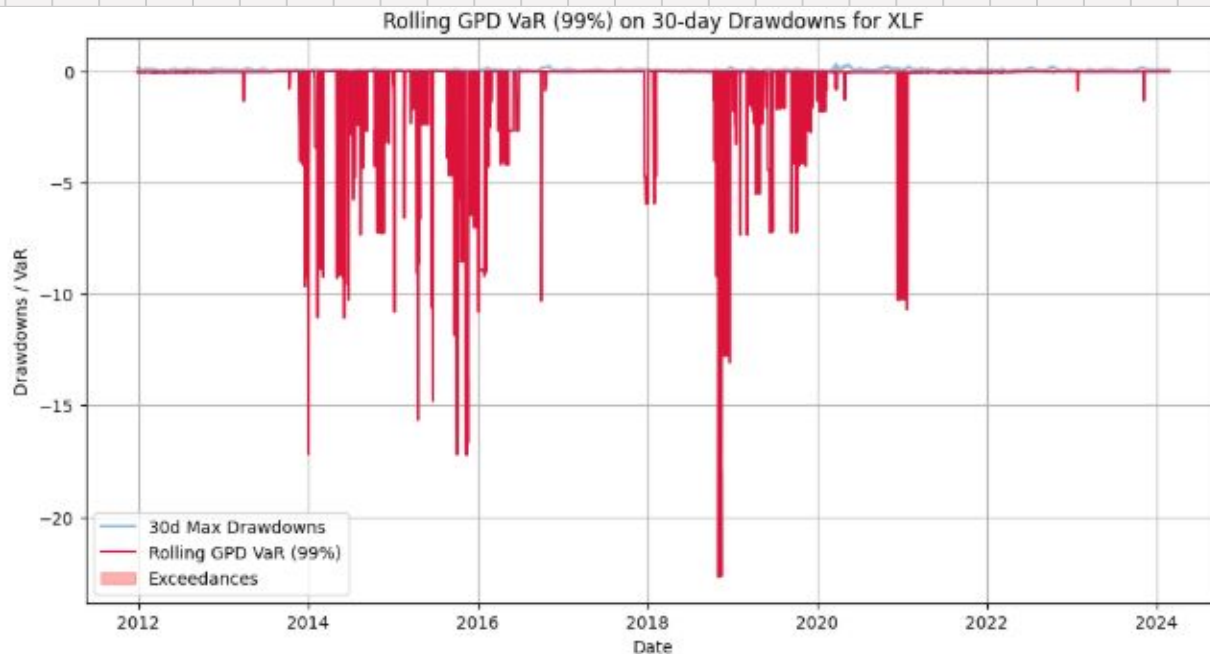


Window = 500
threshold _percentile = 95
P = 0.99

Main results

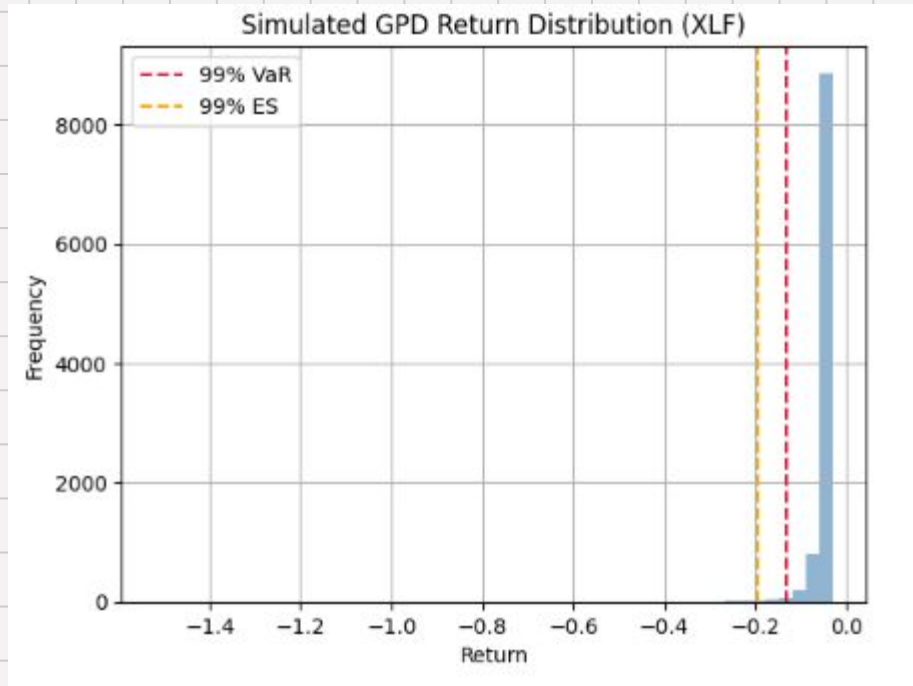
Rolling GPD on drawdowns

★ Kupiec POF Test for Rolling GPD VaR on Drawdowns
Total Observations: 3068
Expected Violations (1%): 30.68
Actual Violations: 16
Observed Violation Rate: 0.0052
LR Statistic: -22.8742



Main results

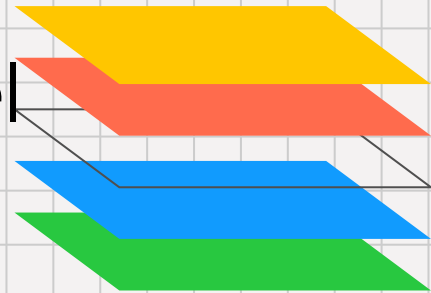
Static GPD + Monte Carlo simulation



Monte Carlo Simulated GPD Tail Risk:
99% VaR: -0.1303
99% Expected Shortfall: -0.1950

Autoregressive Extreme Value model

AEV



- Inspired from Feng et al:
 - Proposed AEV to capture time-varying extreme risk in commodity and steel sector returns
 - They showed that tail losses cluster over time, especially during dynamic shocks- something GPD fails to capture
 - AEV introduces autoregression into the tail process:

$$y_t = \mu + \phi \cdot y_{t-1} + \varepsilon_t, \varepsilon_t \sim \text{GPD}(\xi, \beta)$$

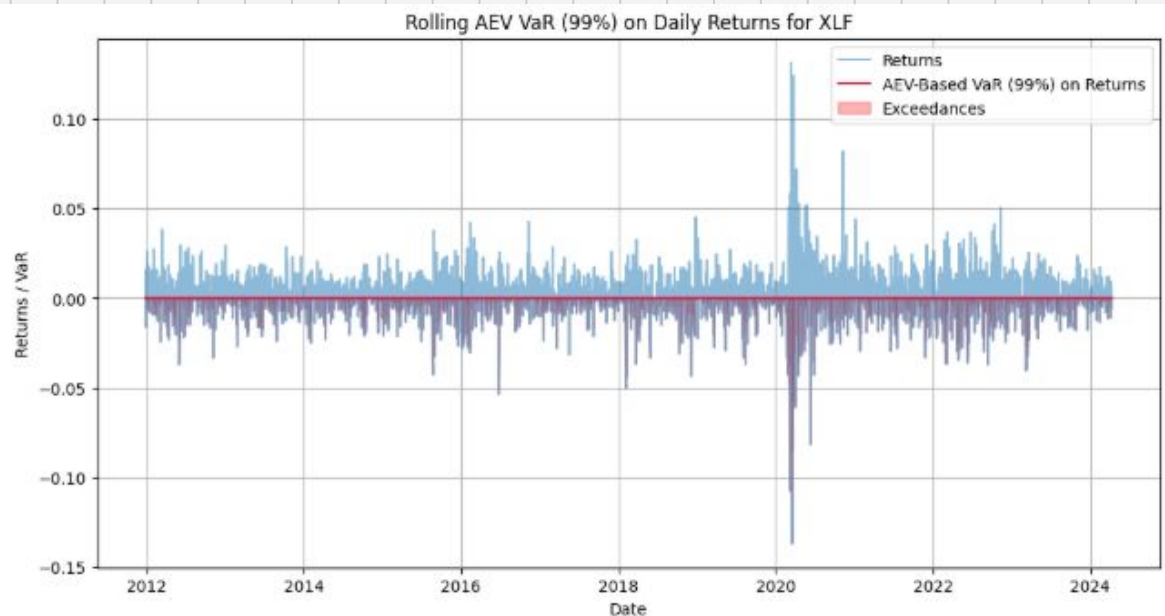
Autoregressive Extreme Value model

Aspect	Static GPD	Rolling GPD	AEV Model
Time Variation	✗ None	✓ Rolling windows	✓ Autoregressive
Tail Clustering	✗ Ignored	⚠ Local only	✓ Modeled directly
Exceedance Independence	✓ Assumed	✓ Assumed	✗ Broken (modeled)
Tail Shock Memory	✗ None	⚠ Short-term	✓ Long-term

Main results

AEV Returns

📌 Kupiec POF Test for AEV on Returns
Total Observations: 3089
Expected Violations (1%): 30.89
Actual Violations: 1460
Observed Violation Rate: 0.4726
LR Statistic: -22.9120

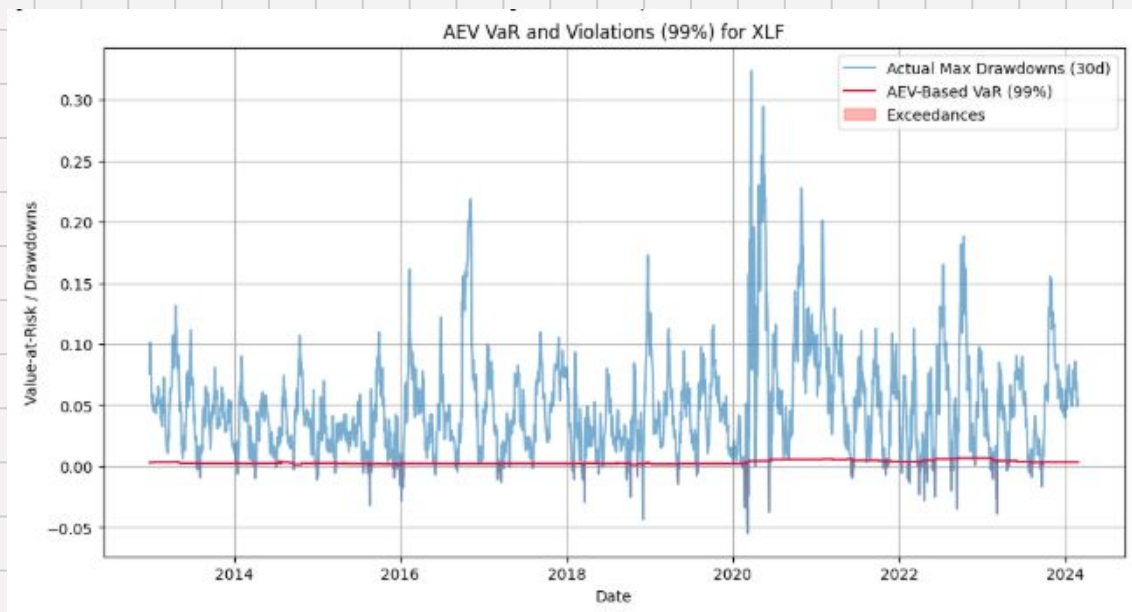


Window = 500
Tail_fraction = 0.1

Main results

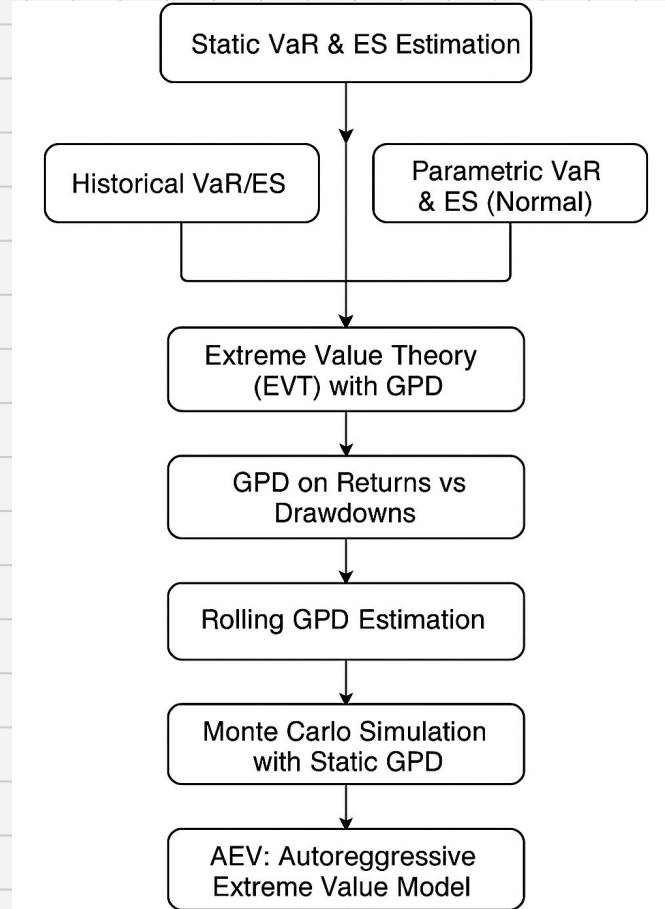
AEV Drawdowns

Window = 500
Tail_frac = 0.1



★ Kupiec POF Test Results
Total Observations: 3060
Expected Violations (1%): 30.60
Actual Violations: 119
Observed Violation Rate: 0.0389
LR Statistic: -22.8742

So
far...



...this is
what
we
have
done

Summary

What models work best?
Under what
circumstances?

-  Best for real-time and stress-aware: Rolling GPD on returns
-  Best for stress simulations: Static GPD + Monte Carlo
-  AEV (both forms): conceptually strong, but currently underperform due to **excess violations** → needs tuning or regime-specific calibration
-  Rolling GPD on drawdowns: **capital inefficient**, but great for **stress testing**
-  Static models: use only as baseline, not for actual risk control

Model Ranking: When to Use Which Tail Risk Model

Model	Tail Fit Accuracy	Adaptive to Regime Change	Stress Testing Use	Risk Bias: Efficiency	Best Use Case Scenario
Rolling GPD (returns)	✓	Moderate (slightly overestimated)	Yes	Conservative (fewer violations)	Good for ongoing portfolio monitoring
Rolling GPD (drawdowns)	✓	Limited adaptivity	Very useful	Overestimated	Use for regulatory buffers or long-term
Static GPD + Monte Carlo	✓	None	Excellent	Risky under stress	Ideal for scenario analysis or capital
AEV (returns)	✗	Long-term dependent memory	Risky under stress	Intended for real-time	Conceptually appealing but needs tuning for returns
AEV (drawdowns)	✗	Long-term Tail clustering model	Intended for real-time	Underestimates risk (high violations)	May fail under volatility spikes or crisis
Static Student-t (baseline)	A	Static	Mildly helpful	Underestimates breaks under stress	Quick benchmark when volatility is stable
Static Normal (baseline)	A	Conservative (fewer violations)	Fails in extremes	Dangerous under coverage	Avoid in any tail-risk context

Recommendations

- Replace static thresholds with dynamic, adaptive models in volatile markets
- Leverage drawdown-based EVT for regulatory stress testing and capital planning
- Explore and refine autoregressive EVT models (e.g., AEV) for non-stationary tail risks

Future Work

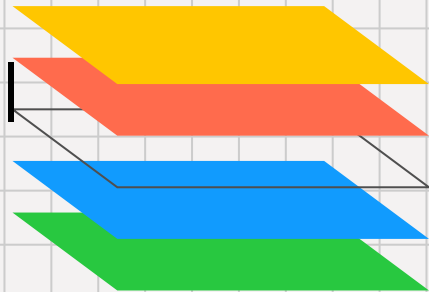
- Refine AEV parameter tuning to improve performance under crisis regimes
- Extend to multi-asset portfolios or sector-based modeling (e.g., financials vs. tech)
- Explore hybrid models combining AEV + GPD with machine learning for structural breaks

Any questions?



Autoregressive Extreme Value model

Adaptation



- Applied the model to XLF, a single financial sector ETF.
- Modeled AEV on returns and drawdowns to capture evolving tail behavior.
- Goal: assess whether tail dependence improves forecasting of extreme risk.